

Token Economics

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Cost per token at scale

PROFESSIONAL DEFINITION

Token Economics analyses the cost of operating large language models on a per-token basis, revealing how prompt design, context length and volume drive operating cost at scale.

WHO DEVELOPED IT

An operational framing from LLM-application practice, formalised as providers priced models per input and output token.

WHY IT WAS DEVELOPED

It was developed to make the recurring cost of LLM applications visible and manageable, since small per-token costs compound enormously at scale.

FIGURE 1 · TOKEN ECONOMICS



Figure 1. A schematic of the Token Economics as applied in BARBM312 (illustrative).

HOW IT IS USED

Cost per token is multiplied across realistic volumes; prompt and context efficiency are optimised to control spend.

WHEN IT IS USED

When budgeting and optimising LLM-based products.

BY WHOM IT IS USED

Product, engineering and finance teams running LLM applications.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It translates abstract model usage into a concrete, controllable unit cost, enabling FinOps discipline for generative AI.

RELATED FRAMEWORKS IN THIS COURSE

AI Value Stack · Slibrary 9

Compute Cost Pyramid · Slibrary 9

Hyperscaler Dependency Map · Slibrary 9

Capex vs. Opex Shift · Slibrary 9

SOURCES (HARVARD REFERENCING STYLE)

OpenAI (2023) 'Pricing' documentation.

Chen, L. et al. (2023) 'FrugalGPT: How to use large language models while reducing cost', arXiv:2305.05176.